

Htet Myat Aung (Alexander E. Hong)

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Education

King Mongkut's University of Technology Thonburi (KMUTT) , Automation Engineering • First Class Honors • Thesis: Natural Gas Methane Leak Detection Using an Infrared Camera • INC Department Outstanding Student Scholarship • Winter Semester 2023 & 2024	Bangkok, Thailand 2022 – 2025
Yangon Technological University (Center of Excellence) , Mechatronics Engineering • Participated in Myanmar RoboLeague 2019 as Mentor • Led First Global Challenge 2019 team representing Myanmar	Yangon, Myanmar 2015 – 2022

Experience

Maintenance Technology Center (MTC) - Non-Destructive Detection Technology Lab , Full Time Machine Learning Researcher Developing advanced machine learning solutions for non-destructive testing applications under Prof. Cherdpong Jomdecha, focusing on acoustic emission sensor data analysis and anomaly detection. • Designed and built efficient data pipelines and workflows using Python and improved data processing efficiency, scalability, and reliability by optimizing pipeline workflow, resulting in faster model training and deployment • Implementing unsupervised clustering and anomaly detection algorithms for pattern recognition in complex time-series sensor data from industrial pipe inspection systems • Contributing to innovative maintenance technology solutions for industrial applications	Bangkok, Thailand Sept 2025 – present 6 months
Maintenance Technology Center (MTC) - Non-Destructive Detection Technology Lab , Machine Learning Intern Independently designed and implemented unsupervised machine learning solutions for GRP pipe inspection using acoustic emission sensor data, demonstrating strong problem-solving and analytical skills in identifying patterns from complex, real-world infrastructure data on Linux based WSL2 platform with version control via Git. • Won First Runner-Up Award in 'Smart Infrastructure for Smart City' competition by Metropolitan Waterworks Authority • Listed in KMUTT Hall of Fame for outstanding achievement in Smart Infrastructure development • Applied AI/ML techniques to solve real-world infrastructure challenges	Bangkok, Thailand June 2025 – Aug 2025 3 months

HoneyHunt.AI, Junior Machine Learning Intern / Large Language Model Intern
Enhanced automated resume parsing and intelligent job matching systems by integrating AI APIs (advanced LLM APIs), improving accuracy and scalability.

Bangkok, Thailand
Dec 2024 – Feb 2025
3 months

- Improved algorithm scalability and matching accuracy for production-level deployment
- Contributed to product development in AI-powered recruitment technology
- Explored Generative AI techniques to design agentic workflows for automated data processing and decision-making, leveraging LLMs and prompt engineering to optimize AI-driven recruitment solutions
- Implemented API management best practices including versioning, rate limiting, and authentication using API gateways to support reliable deployment of GenAI workflows in production environments

Research

KMUTT - OPTIMISED 3D-CNN FOR REALTIME INFRARED NATURAL GAS LEAK CLASSIFICATION, Computer Vision (Student) Researcher

Bangkok, Thailand
June 2024 – Sept 2025

Developed optimized 3D CNN architecture for real-time infrared video classification achieving 99.75% accuracy in methane gas leak detection, processing temporal-spatial features from thermal camera streams.

- Implemented classical image processing preprocessing pipeline including Gaussian Mixture Models (GMM), Running Average, and adaptive background subtraction for robust foreground segmentation in varying lighting conditions
- Optimized model for edge deployment through input resolution scaling (half/quarter size variants) and quantization techniques, maintaining 99%+ accuracy while reducing inference time by 60%
- Co-Winner of Best Paper Award at 13th International Conference on Information and Communication Technology Bandung (Hybrid)
 - IcoICT 2025

CODIA Lab, KMUTT & Panasonic Singapore, Perception (Student) Research Engineer

Bangkok, Thailand
Feb 2024 – July 2024

Developed and optimized SLAM Sparse Bundle Adjustment algorithms in C for visual odometry applications for real-time edge deployment on ARM-based embedded platforms.

- Optimized SLAM Sparse Bundle Adjustment algorithms for edge devices, improving computational efficiency and enabling real-time performance in resource-constrained environments

COSIN Lab, INC Department, Student Researcher

Bangkok, Thailand
Feb 2024 – July 2024

Developed ROS2-based autonomous navigation systems integrating IMU and camera sensor fusion for competitive robotics applications.

- Implemented micro-ROS framework on microcontrollers for real-time IMU data acquisition and sensor synchronization, achieving sub-10ms latency for control loop feedback
- Performed sensor calibration including IMU bias estimation and camera extrinsic alignment to improve localization accuracy in dynamic environments

Volunteer

FIRST GLOBAL CHALLENGE 2019, Mentor

Dubai, UAE
June 2019 – Oct 2019

Provided technical guidance and strategic mentorship to Myanmar national robotics team, fostering knowledge sharing through structured training sessions and collaborative problem-solving.

- Led team to win Judges Award at international robotics competition
- Fostered STEM education and international collaboration in robotics

Laboratory of Robotics and Artificial Intelligence (L.O.R.A), Robotics Researcher and Founder

Founded and led comprehensive robotics research laboratory focusing on practical applications.

- Spearheaded development of autonomous disinfection robots during COVID-19 pandemic in Myanmar
- Managed multiple concurrent research initiatives and mentored junior researchers
- Managed cross-functional communication between engineering teams, public health stakeholders, and external partners to align technical development with societal needs during the pandemic

Yangon, Myanmar
Dec 2019 – June 2021

ConceptX International School, Head of Department and Instructor of Robotics and Computer Science

Instructed Computer Science and Robotics to approximately 200 students from Upper Primary 5 to Lower Secondary 2.

- Promoted to Head of Department in October 2021
- Taught IGCSE Computer Science to Secondary 3 students

Yangon, Myanmar
June 2021 – Dec 2021

Awards and Honors

Best Paper Award - IcoICT 2025 (IEEE Xplore), 13th International Conference on Information and Communication Technology, Bandung
Co-winner with Asst. Prof. Dr. Sarawan Wongs

July 2025 – present

Best Presentation Award - ICA-SP Con 2025, 16th Instrumentation, Control and Automation Senior Project Conference

Apr 2025 – present

First Runner-Up - Smart Infrastructure for Smart City Competition, Metropolitan Waterworks Authority, Bangkok

July 2025 – present

KMUTT Hall of Fame, King Mongkut's University of Technology Thonburi

July 2025 – present

INC Department Outstanding Student Scholarship, KMUTT - Winter Semester 2024

2024 – present

INC Department Outstanding Student Scholarship, KMUTT - Winter Semester 2023

2023 – present

Judges Award - First Global Challenge 2019, International Robotics Competition, Dubai, UAE

Oct 2019 – present

Publications

Optimized 3D-CNN for Real-Time Infrared Natural Gas Leak Classification

Developed a high-accuracy (99.75%) 3D CNN model for methane gas leak detection using infrared imaging. Winner of Best Paper Award. DOI - 10.1109/ICoICT66265.2025.11192973
Htet Myat Aung, Sarawan Wongs
ieeexplore.ieee.org/document/11192973

Natural Gas Methane Leak Detection Using an Infrared Camera

Best Presentation Award winner for research on natural gas leak detection systems using infrared imaging technology.
Htet Myat Aung

Skills

Perception & Vision

Robotics

Programming Languages

ML/DL

Tools

Languages

English

Fluent (TOEFL 84 - Writing: 25, Speaking: 26 & Extensive technical documentation and international collaboration experience)

Burmese

Native

Interests

Artificial Intelligence & Machine Learning

Robotics & Automation

Research & Innovation